

EE1101: Circuits and Network Analysis - Quiz 5

Name: _____

Roll No: _____

Instructions

1. The exam duration is 30 minutes, starting at 15:40. Ensure that you return the paper by 16:10. Please be aware that **late submissions** may incur **penalties**.
2. Go through all the questions carefully. Any misinterpretations will **not** be considered. Write your solutions in the designated space provided for each question. Answers written outside these areas will not be accepted.
3. Adopting unfair practices will lead to an F grade and will be reported to the disciplinary committee.

I hereby declare that I have read all the instructions carefully and will adhere to them.

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1. (4 points) Compute the inverse hybrid parameters of the two port network shown in Fig. 1.

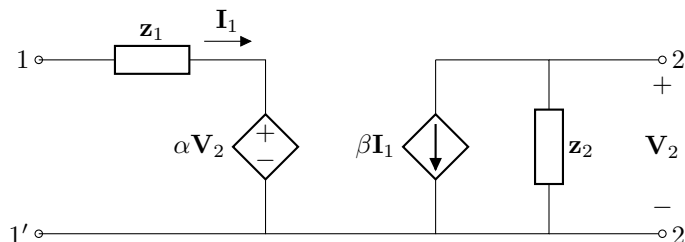


Fig. 1: Two port network for question 1

- (1 point) Write the parameter $\mathbf{g}_{11}$ here: _____.
 - (1 point) Write the parameter $\mathbf{g}_{12}$ here: _____.
 - (1 point) Write the parameter $\mathbf{g}_{21}$ here: _____.
 - (1 point) Write the parameter $\mathbf{g}_{22}$ here: _____.
2. (4 points) For the circuit shown in Fig. 2, compute the Norton equivalent circuit as seen from the port $a-o$. The two-port network is characterized by the admittance parameters $\mathbf{y}_{11}$, $\mathbf{y}_{12}$, $\mathbf{y}_{21}$ and $\mathbf{y}_{22}$.
 - (2 points) Write the expression for Norton Current $\mathbf{I}_n$ here: _____.
 - (2 points) Write the expression for Equivalent impedance $\mathbf{z}_n$ here: _____.
 3. (2 points) Consider the circuit shown in Fig. 3. If a variable resistor (say R_l) is connected at the port 2-2', compute
 - (1 point) Compute the value of R_l at which maximum active power can be extracted. Enter your answer here: _____.

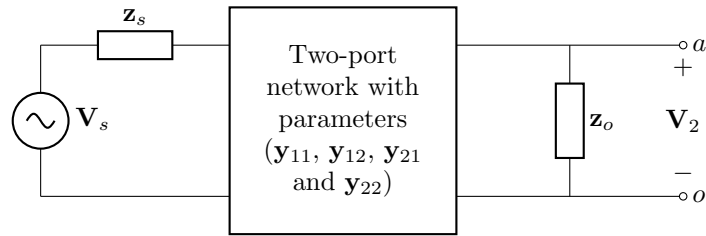


Fig. 2: Circuit for Question 2

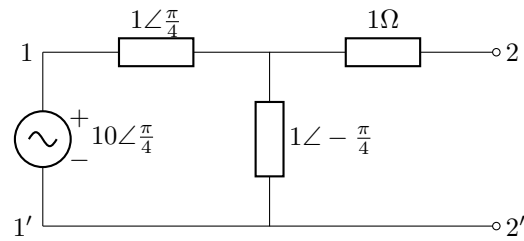


Fig. 3: Circuit for question 3

- (b) (1 point) Compute the voltage across the port 2-2' when maximum active power is extracted. Enter your answer here: _____.

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1. (4 points) Compute the impedance parameters of the two port network shown in Fig. 1.

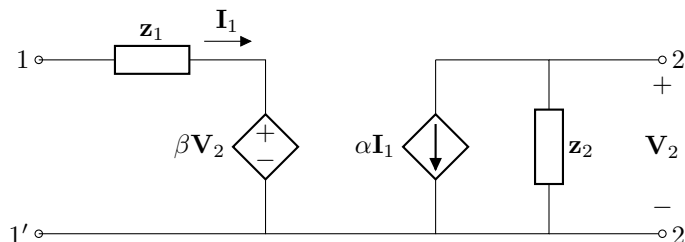


Fig. 1: Two port network for question 1

- (1 point) Write the parameter $\mathbf{z}_{11}$ here: _____.
 - (1 point) Write the parameter $\mathbf{z}_{12}$ here: _____.
 - (1 point) Write the parameter $\mathbf{z}_{21}$ here: _____.
 - (1 point) Write the parameter $\mathbf{z}_{22}$ here: _____.
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 - (2 points) Write the expression for Norton Current $\mathbf{I}_n$ here: _____.
 - (2 points) Write the expression for Equivalent impedance $\mathbf{z}_n$ here: _____.
 3. (2 points) Consider the circuit shown in Fig. 3. If a variable impedance (say $\mathbf{z}_l = R_l + jx_l$) is connected at the port 2-2', compute
 - (1 point) Compute the value of R_l at which maximum active power can be extracted. Enter your answer here: _____.

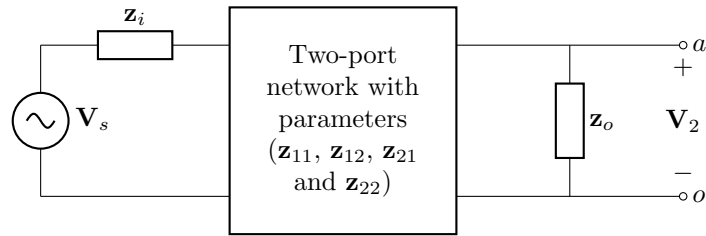


Fig. 2: Circuit for Question 2

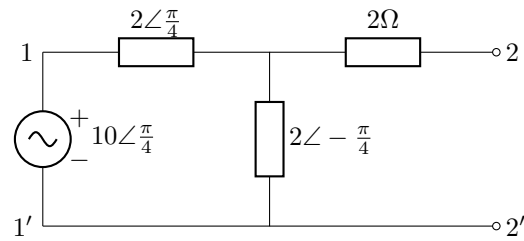


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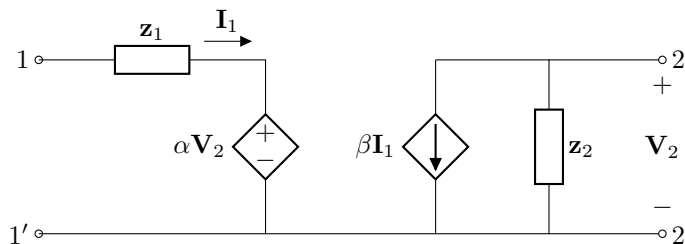


Fig. 1: Two port network for question 1

- (1 point) Write the parameter z_{11} here: _____.
 - (1 point) Write the parameter z_{12} here: _____.
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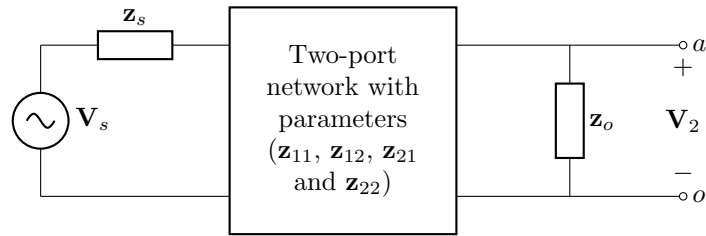


Fig. 2: Circuit for Question 2

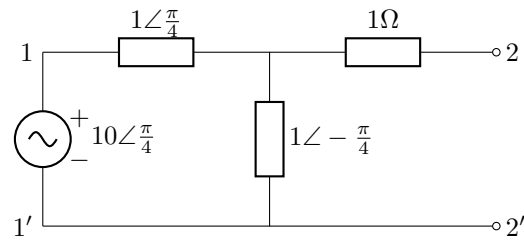


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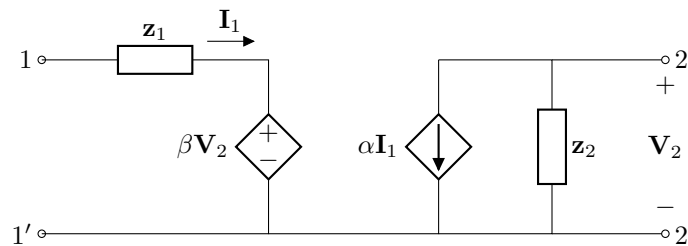


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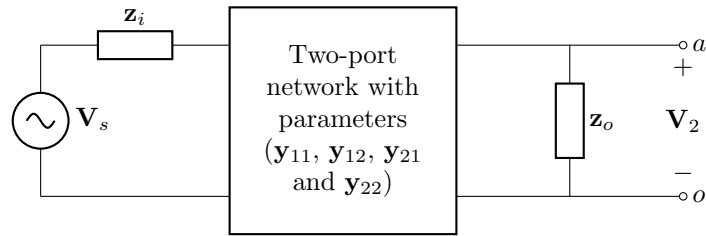


Fig. 2: Circuit for Question 2

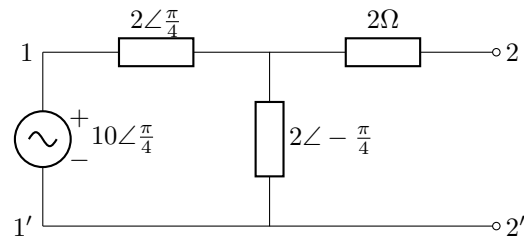


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